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SUPERSEDING
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FEDERAL SPECIFICATION

SWEEPERS, ROTARY, SELF-PROPELLED INDUSTRIAL TYPE, RIDER-OPERATED

This specification was approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers gasoline, diesel, liquified petroleum-gas (LPG) engine powered, and battery powered, commercial, self-propelled, rider operated, industrial type sweepers equipped with a vacuum dust control system and hopper to collect and transport sweepings.

1.1.1 Federal specification coverage. Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification. Sweepers shall be one of the following types and sizes, as specified (see 6.2):

Type I - Gasoline- or diesel-engine-driven
Type II - Electric-motor-driven, battery powered (sizes 36 and 42 only)
Type III - LPG engine driven

Size 28 - Main brush length not less than 28 inches
Size 36 - Main brush length not less than 36 inches
Size 42 - Main brush length not less than 42 inches
Size 48 - Main brush length not less than 48 inches

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications

W-B-131 - Battery, Storage, Vehicular, Ignition, Lighting, and Starting
W-B-133 - Batteries, Storage (Lead Acid, Industrial Portable Service)

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VV-G-1690 - Gasoline, Automotive, Leaded or Unleaded
ZZ-T-391 - Tire, Solid Rubber, and Wheels, Solid Rubber Tire, (Industrial)
ZZ-T-410 - Tire Pneumatic Industrial

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards, and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Single copies of this specification and other Federal specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.)

(Federal Government activities may obtain copies of Federal standardization documents, and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specifications

MIL-V-173 - Varnish, Moisture, and Fungus Resistant (For Treatment of Communications, Electronic, and Associated Equipment)
MIL-T-704 - Treatment and Painting of Material

Military Standards

MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment
MIL-STD-461 - Electromognetic Emission and Susceptibility Requirements For the Control of Electromagnetic Interferance

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Federal Regulations

DEPARTMENT OF TRANSPORTATION (DoT)

Federal Motor Vehicle Safety Standards and Regulations
Motor Carrier Safety Regulations

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

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ENVIRONMENTAL PROTECTION AGENCY (EPA)

Code of Federal Regulations, Title 45, Part 1201 - New Motor Vehicles and Engines

(Application for copies should be addressed to the Public Affairs Office, Environmental Protection Agency, Rockville, MD 20852.)

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

STATE OF CALIFORNIA

California Vehicle Code

(Application for copies should be addressed to the Department of Motor Vehicles, 2570 24th Street, Sacramento, CA 95809.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

MG - 1 - Motors and Generators

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Description. The sweeper shall be a power-driven wheeled vehicle consisting of a welded steel frame, driven brushes, a hopper to collect sweepings, a vacuum dust control, automotive type steering, brakes, operating controls, and all other parts and accessories necessary for the purpose intended.

3.2 First article. When specified (see 6.2), the contractor shall furnish a complete sweeper for first article inspection and approval (see 4.2.1 and 6.2).

3.3 Repair parts and service. Replacement, or repair, parts, and service for the product provided under this specification shall be available from the manufacturer's regular, or through commerical parts distribution, or services organizations.

3.3.1 Identification or model numbers. The product provided under this specification shall be identifiable by the manufacturer's regular or by commercial parts distribution, or service organizations.

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.5 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.6 Construction. The equipment shall be designed and constructed to facilitate field maintenance. All adjustments and replaceable accessories shall be readily accessible. Conditions which can be hazardous to personnel or deleterious to equipment under normal use shall not be permitted.

3.6.1 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. When specified (see 6.2), the unit shall be cleaned, treated, and painted in accordance with MIL-T-704, type A. In either case, color of the finish coat shall be as specified (see 6.2).

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3.6.1.1 Cleaning. Metal surfaces to be painted shall be cleaned to insure that they are free from all oil and grease, welding slag and spatter, all mill scale, which can be removed by power wire brushing, products of corrosion, dirt, or other foreign substances.

3.6.1.2 Treatment. As soon as practicable after cleaning and before any evidence of rust or other contamination can result, metal surfaces shall be treated with a primer pretreatment coating to increase the adhesion of the coating system.

3.6.1.3 Painting. Painting shall consist of at least one coat of rust-inhibiting primer and one coat of finish enamel. The rust-inhibiting primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current process except that the total dry film thickness shall be not less than 2.5 mils. The paint shall be free from runs, sags, orange peel, or other defects.

3.6.2 Identification marking. Each sweeper shall bear the manufacturer's name or trademark on a nameplate securely affixed in a conspicuous place. In lieu of the nameplate, the manufacturer's name or trademark may be cast integral with, stamped, or otherwise permanently marked upon the sweeper. An identification plate will be furnished by the contracting officer for each sweeper. The contractor shall stamp all necessary data in the blank spaces of the plate provided for the purpose, and securely affix a plate to each sweeper in a conspicuous place. The applicable nomemclature contained in the contract item description shall be placed in the top blank.

3.6.3 Instruction plates. The sweeper shall be equipped with instruction plates or decals suitably located, describing any special or important procedures to be followed in operating and servicing the sweeper or warnings of hazardous procedures. Plates shall be of a material which will last and remain legible for the life of the sweeper, and shall be securely affixed thereto. Properly affixed plastic decals are also permitted.

3.6.4 Fungus resistance. When specified (see 6.2), electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173, except that:

a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
b. Current-carrying contact surfaces, such as relay contact points, shall not be coated.

3.6.5 Electromagnetic interference control. When specified (see 6.2), the sweeper shall conform to the electromagnetic radiation suppression characteristics of SAE J552, or electromagnetic interference control requirements of MIL-STD-461 for class IIIC equipment as specified (see 6.2).

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3.6.6 Lifting and tying down attachments. When specified (see 6.2), the sweeper shall be equipped with lifting and tying down attachments. Lifting and tying down attachments shall conform to MIL-STD-209, type III, except a 3-point helicopter lift is acceptable. Attachments shall be indicated by a transportation plate, except tying down attachments may be identified by stenciling or other suitable marking on the sweeper. Tying down marking shall clearly indicate that the attachments are intended for tying down of the sweeper on the carrier when shipped.

3.7 Capacities and operational characteristics. Sweepers shall conform to the requirements of table I.

TABLE I. Capacities and operational characteristics

	Size			
	28	36	42	48
Length of main brush, not less than (inches)	28	36	42	48
Diameter of side brush, not less than (inches)	14	19	21	21
1/ Engine horsepower (net), not less than	7	12	18	25
Sweeping width (swath) with one side brush, not less than (inches)	40	47	53	60
Rated hopper capacity, not less than (cubic feet)	4.5	9	12	14
Maximum sweeping speed, not less than (miles per hour)	5	5	5	9
Maximum outside turning radius; not more than (inches)	65	76	88	90

1/ Engine horsepower (hp) (net) shall be equal to the dynamometer rating of the engine, complete with cooling fan, muffler, generator or alternator, and air filter at not higher than 3,6000 revolutions per minute when corrected to 29 inches of mercury barometric pressure and 60 degrees Fahrenheit (°F) ambient temperature.

3.8 Performance. Sweepers shall be capable of operating in ambient temperature ranging from +125°F to -10°F. Sweepers shall effectively sweep and retain dust, dirt, and debris while operated at the sweeping speed recommended by the contractor, along hard surfaced areas, gutters, and ramps. Types I and III sweepers shall be capable of ascending and sweeping a 20 percent grade, and type II sweeper shall be capable of ascending and sweeping a 12 percent grade. When tested as specified in 4.4.2, the sweeper shall, in a one-way pass, pick up and retain not less than 95 percent of the sand while traveling at the manufacturer's recommended sweeping speed, but not less than 2-1/2 miles per hour (mph), except for the size 28 sweeper which shall be tested at not less than 1-1/2 mph. No visible dust shall escape from the vacuum system. When tested as specified in 4.4.3, the sweeper shall, in a one-way pass, pick up and retain not less than 90 percent of the specified items while traveling at not less than 5 mph. There shall be not be more than two items of the same kind in any group of items not picked up. The sweeper shall show no evidence of deformation, fracture, or failure of any part or

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component. The sweeper shall be capable of passing over metal obstructions extending not less than 2 inches above a level surface while sweeping at the sweeping speed specified in table I, without damage to any part of the sweeper. The brushes, vacuum fan, and hopper dump shall each be independently operable.

3.9 Safety provisions. All moving parts, parts subject to high temperature, and electrical components that are of such nature or so located as to become a hazard to operating personnel, shall be insulated, fully enclosed, or guarded. Batteries shall be so located that electrolyte leaking from defective or damaged cells will not contact operating personnel or gasoline tanks and liners. Operator's platform and foot operated controls shall be provided with nonskid surfaces. Operator shall have a clear and unobstructed view in all directions.

3.10 Air pollution control. The type I and type III sweepers shall comply with the applicable EPA Regulations governing Control of Air Pollution for New Motor Vehicles and New Motor Vehicle Engines in effect on the date of manufacturer, unless the sweepers are specifically exempted or excluded from these regulations. In addition, vehicles destined for end use in California shall comply with State of Califiornia regulations governing air pollution control in effect on the date of manufacture unless specifically exempted or excluded from these regulations.

3.11 Noise limits. The noise level of the sweeper assembly at the operator's position shall not exceed 92 decibels (A) with brush and vacuum on, with speed not exceeding 3 mph, when tested in accordance with 4.4.9.

3.12 Identical items. All units of the same classification furnished under a specific contract shall be physically and mechanically identical. Parts, accessories, assemblies, and components are included in this requirment. Written approval for deviations must be obtained in advance from the contracting officer.

3.13 Components.

3.13.1 Frame. Frame shall be fabricated from welded steel not less than 0.1345-inch (U.S. revised standard gage No. 10). Frame shall withstand distortion and damage under the maximum loads and stresses encountered in sweeping operations.

3.13.2 Housing. The main brush and brush mounting components shall be enclosed in a sturdy sheet metal housing. The sides and rear of the housing shall be equipped with tear-resistant skirts or flaps of not less than 2-ply rubber, 1/4-inch thick neoprene, or synthetic-rubber-impregnated-fabric sufficiently flexible to provide an efficient dust seal between the housing and the surface to be swept. Where required, seals shall be provided to confine the vacuum system airflow. The housing shall be either readily removable from the frame or shall be provided with dustproof doors that permit

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rapid and easy access for the removal, maintenance, and repair of the components within the housing. The housing shall be designed to permit dust, dirt, and debris up to 2-1/2 inches in height to pass into the main brush housing area.

3.13.3 Power system.

3.13.3.1 Type I. The type I sweeper shall be powered by a gasoline or diesel engine as specified (see 6.2). The continuous brake hp and speed rating shall be in excess of that required to meet the operational requirements of the sweeper. The engine shall be complete with the accessories normally furnished with the sweeper, including at least the following:

a. A 12-volt (V) electric system including a starter; battery charging generator or alternator of not less than 15-ampere (A) rating, except that the size 28 sweeper shall have a minimum 10-A rating; regulator and ammeter, and a dry-charged battery without electrolyte.
b. An engine housing.
c. A fuel tank of sufficient capacity for 8 hours of normal operation.
d. Fuel filter located between fuel tank and carburetor.
e. Speed governing system with provisions for adjusting the speed setting.
f. An engine hour meter.
g. A fuel level gage.
h. Air cleaner for air inlet.

3.13.3.2 Engine. The engine shall be air or water cooled, 4-stroke cycle, industrial or automotive type of a commercial design that has been proven to be satisfactory in extensive use and for which repair parts are readily obtainable. The engine for size 42 and 48 sweepers shall have not less than two cylinders.

3.13.3.2.1 Unleaded fuel for gasoline engines. Gasoline engines shall produce the required performance on unleaded gasoline conforming to VV-G-1690. Unless otherwise specified (see 6.2), regular grade gasoline shall be used.

3.13.3.2.2 Diesel engine starting requirements. Diesel engines shall achieve self-sustained combustion at the manufacturer's rated idle speed in less than 5 minutes in ambient temperature from -20°F to +125°F. The engines shall be ready for full load operation in less than 15 minutes.

3.13.3.2.3 Engine battery. The battery shall be a standard 12-V battery, rated for not less than 40-A hours at the 20-hour rate, and be capable of performing not less than three engine starts at ambient temperatures as low as -10°F, and to operate all accessories specified. The battery shall be clamped within replaceable steel brackets and shall conform to W-B-131. Unless otherwise specified (see 6.2), a battery shall be furnished with each sweeper.

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3.13.3.2.4 Lubricating oil filter. The engine lubricating oil system for sizes 36, 42, and 48 sweepers shall include a standard type filter mounted in an accessible location for replacement of the element. The element size shall be that recommended by the filter manufacturer for the particular engine so as to achieve maximum filtration efficiency.

3.13.3.2.5 Crankcase ventilation system. The engine shall be provided with a crankcase ventilation system to purge vapors and relieve pressure. Intake air shall be passed through an air filter prior to entering the crankcase.

3.13.3.3 Type II. The type II sweeper shall be powered by direct current electric motor(s) designed for either 24 or 36 V. Motor(s) shall be of the high starting torque, heavy-duty type conforming to the applicable requirements of NEMA MG1. Continuous rated hp of the motor(s) shall be not less than that required for its particular operation when sweeper is tested as specified in 4.3.1 through 4.3.5.

3.13.3.3.1 Speed control system. The speed control sytem shall regulate the travel from zero to maximum speed and shall permit accurate maneuvering in restricted areas. The sweeper shall be equipped with a selector switch or foot operated control for forward and reverse travel, and an enclosed main switch so wired as to permit complete isolation of the battery. All systems shall be protected against increased temperature from overloading and short circuits.

3.13.3.3.2 Type II Sweeper battery(s). The sweeper shall be designed for operation with either 24 or 36 V, industrial type, lead-acid storage battery(s) conforming to W-B-133, class 1. The battery(s) shall be of not less than 15.0 kilowatt-hour capacity, to permit not less than 6 hours normal sweeping operation.

3.13.3.3.3 Battery compartment. A compartment for battery(s) and boxes shall be constructed to permit quick battery inspection, servicing, and changing. Effective means shall be provided for securing the battery(s) and boxes in place. The compartment cover shall be lined with insulating materials to prevent short circuit of battery terminals if the sweeper is overturned.

3.13.3.3.4 Charging receptacle. A battery charging receptacle shall be provided. Receptacle and wiring shall have an ampere rating of not less than the high charging rate of the battery. Specific types of receptacles and connectors shall be as specified (see 6.2 and 6.4).

3.13.3.4 Type III. The type III sweeper shall be powered by an engine to permit safe and efficient use of LPG as fuel. All components of the LPG fuel system shall be accepted standard commercial products shall be demonstrated to be a safe compatible system. Engine shall be complete with the accessories normally furnished with the sweeper, including at least following:

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a. A 12-V electrical system including a starter, regulator, ammeter, and a dry-charged battery without electrolyte.
b. Engine housing.
c. Speed governing system with provisions for adjusting the speed setting.
d. An engine hour meter.
e. Air cleaner for carburetor inlet.

3.13.3.4.1 Sizes 42 and 48. LPG engines for sizes 42 and 48 sweepers shall be provided with the following additional accessories:

a. A battery charging generator or alternator of not less than 15-A rating.
b. A liquid withdrawal LPG system including a primary and secondary regulator, vaporizer, and carburetor designed specifically for LPG operation.
c. A liquid withdrawal LPG tank of sufficient capacity for not less than 6 hours operation under normal loads. Tank shall be equipped with manual shutoff valve, quick disconnect line coupler, visible sight fuel gage, and mounting bracket with quick release clamps for easy tank replacement.

3.13.3.4.2 Sizes 28 and 36. LPG engines for sizes 28 and 36 sweepers shall be provided with the following additional accessories:

a. A battery charging generator or alternator of not less than 10-A rating.
b. A vapor withdrawal LPG system, including a primary and secondary regulator, and LPG carburetor.
c. A vapor withdrawal LPG tank sufficient capacity for not less than 6 hours operation under normal loads. Tank shall be equipped with manual shutoff valve, quick disconnect line coupler, visible sight fuel gage, and mounting bracket with quick release clamps for easy tank replacement.

3.13.4 Power transmission system. The sweeper shall be equipped with a heavy-duty mechanical or hydraulic drive system to transmit power to the drive wheels either directly or through a final roller chain drive. The drive shall include provisions for forward and reverse travel.

3.13.5 Steering. The sweeper shall be equipped with automotive type steering mechanism with steering wheel, and with sufficient reduction in ratio to allow adequate directional control. Adjustment shall be provided to limit the free play to a maximum of 1/8 of one wheel revolution. Steering wheel shall be not less than 16 inches in diameter to allow steering with minimum effort.

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3.13.6 Wheels and tires.

3.13.6.1 Wheels. The sweeper shall have not less than three wheels. Wheels shall be provided with ball or roller bearings and be demountable at the hub or rim. The weight of the sweeper shall be distributed over the wheels in proportions that will assure proper traction for propulsion and steering under all operating conditions. Rims shall comply with the recommendations of TRA.

3.13.6.2 Tires. Unless otherwise specified (see 6.2), manufacturer's standard tires shall be furnished. Tire size shall be the manufacturer's current standard size for the sweeper specified. Tire capacities shall be not less than the individual wheel loadings imposed by the fully loaded sweeper and shall be in accordance with the recommendations of TRA. When tube type tires are furnished, inner tubes shall be heavy-duty type and shall be of proper size for tires. When specified (see 6.2), tires shall be tube, tubeless, or solid type with nondirectional mud and snow tread, nondirectional cross-country tread, or open diamond tread which will not pick up stones or foreign objects in conformance to ZZ-T-391 and ZZ-T-410, as required.

3.13.7 Brakes. The sweeper shall be equipped with foot controlled service brakes on not less than two wheels. Parking brakes control shall be provided with positive lock. Brakes shall be protected against the entry of dirt, grease, and water. Service brakes shall safely stop the sweeper from a 5 mph speed, when traveling forward down a 20 percent grade, dry, paved ramp, within a distance of 7 feet with the hopper loaded with dry sand in the minimum amount as follows: size 28 - 250 pounds (lb); size 36 - 500 lb; size 42 - 750 lb; size 48 - 1,000 lb. Parking brake shall be capable of holding the sweeper on a 20 percent grade, dry paved ramp, while heading either up or down ramp, without evidence of slippage. Either braking system shall be capable of producing emergency stops without causing loss of control. When hydraulic drive system is provided, the braking ability shall conform to the requirements specified herein.

3.13.8 Lighting system. The sweeper shall be equipped with an electric lighting system, including not less than two headlights, one spotlight for each side brush, a combination taillight and stoplight, and an instrument panel light.

3.13.9 Horn. An electric-actuated horn shall be provided. The sound level of the horn shall be greater than the sound level of the sweeper operating at full power.

3.13.10 Dust control system. The dust control system shall be of the vacuum type and shall include the housing with skirt, a suction fan, dust filter, seals, ducts, and controls. Parts shall be constructed of tough and durable materials that provide long wear. Connections shall remain airtight when subjected to continuous vibration from sweeper operations. The system shall meet the test requirements specified in 4.4.5.

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3.13.10.1 Suction fan. The suction fan shall move air from the main brush compartment through the filter in sufficient volume during sweeping operations to prevent escape of visible dust from brush compartment directly to the atmosphere. The suction fan shall operate at a constant speed regardless of sweeper travel speed.

3.13.10.2 Filter. The filter shall be of the dry type and shall have a capacity in balance with the output of the suction fan. Filter bags, when used, shall effectively filter dust from the air before the air is exhausted to the outside. All filters shall be replaceable, made of tear resistant natural or synthetic material, and protected from damage during operation. Power cleaning of the filter by motor, electric vibrator, or their power means shall be provided.

3.13.11 Sweeping system. The sweeping system shall consist of a main brush, side brush or brushes, and a hopper. Conveniently located controls shall be furnished that permit the operator to raise the brushes clear from the surface to eliminate brush drag and wear when traveling. The swath overlap between main brush and side brush shall be such that machine sweeps cleanly without dirt streaks between brush swath. The outer edge of the side brush shall be visible to the driver to allow safe operations while sweeping close to obstructions. Brushes shall be mounted on sealed ball or roller bearings. The system shall incorporate floating and adjustable position brush holders.

3.13.11.1 Main brush. The main brush shall be supported in a carrier having provisions for adjusting the brush vertically from operator's position to any desired amount of floor contact. When specified (see 6.2), the brush shall be driven at a constant speed regardless of the sweeper travel speed. Brush or brush strips shall be exchanged in not more than 20 minutes with standard handtools. Brush shall be not less than 14 inches in diameter. Bristles shall be not less than 3 inches in length. Unless otherwise specified (see 6.2), bristle material shall be natural fiber, synthetic fiber, or a combination thereof.

3.13.11.2 Side brush. Unless otherwise specified (see 6.2), the sweeper shall be equipped with a right side brush. When specified (see 6.2), the sweeper shall be equipped with both right and left side brushes. Side brush shall be adjustable for sweeping position and wear. Brush shall be capable of removing dirt and debris from along walls and curbs by sweeping the material into the path of the main brush. The brush mounting shall provide for flexibility to avoid impact damage, and to allow the brush to follow irregularities in the sweeping surface. Brush and mounting shall be guarded by the frame or a bumper against fixed objects. Side brush mounting shall be designed to be quickly and easily removed by use of standard handtools. Unless otherwise specified (see 6.2), brush bristles shall be of synthetic fiber. When specified (see 6.2), the side brush shall be hydraulically driven at constant speed regardless of sweeper travel speed.

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3.13.11.3 Hopper. The hopper shall be of sheet steel construction, designed to receive, hold, transport, and dump sweepings. The hopper shall be sealed against dust leakage and shall be capable of self-loading and retaining sweepings up to 80 percent of applicable rated capacity specified in table I. With the hopper loaded in this manner, and the brushes turned off and raised to traveling position, the sweeper shall be capable of being driven 100 feet with no visible spillage of material. A power operated means shall be provided for dumping the hopper to remove debris from sizes 36, 42, and 48 sweepers. The hopper of the size 28 sweeper shall be dumped by manual, hydraulic, or power operated means.

3.13.12 Hydraulic system. The hydraulic system shall include rams, pump, reservoir, valves, fittings, and all other parts essential to efficient operation. Hydraulic propelling system, when used, shall in addition, incorporate hydraulic motor(s), flow control valve, and means to keep the oil within operational temperature.

3.13.13 Bearings and dirt seals. All rotating, highly loaded, or high speed parts shall be provided with ball or roller bearings. Antifriction bearings that are sealed and lubricated for life shall be used in areas exposed to extreme dirt and dust.

3.13.14 Operator's seat and controls. The sweeper shall have a comfortable and conveniently located operator's seat. Seat and back shall be cushioned and covered with a durable water-resistant material. All controls necessary for safe and efficient operation shall be furnished in a location convenient to the driver. Controls regulating the forward or reverse motion of the sweeper, including clutches, hydraulic and electric speed controllers, and service brakes shall be foot operated. Operator's seat of type II sweeper shall be equipped with an automatic switch actuated by the operator's weight, or with other suitable means, to stop and lock the brakes on the sweeper when the operator's weight is removed.

3.14 Lubrication. Lubrication means shall be provided for all parts of the equipment normally requiring lubrication. Where the use of high lubrication pressure will damage grease seals or other parts, fittings with pressure release shall be used. Sweeper shall be lubricated prior to delivery. The sweeper shall be conspicuously tagged to identify the lubricants and their temperature range.

3.15 Fire extinguisher. When specified (see 6.2), a fire extinguisher, as specified by the procuring actiivty shall be provided and mounted vertically in a location readily accessible to the operator.

3.16 Workmanship. There shall be no defect that affects serviceability or appearance.

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3.16.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.16.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.16.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.16.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.16.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

3.17 Safety standards. Sweepers and furnished accessories shall comply with all applicable DoT Federal motor vehicle safety standards and regulations in effect on the date of manufacture.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one sweeper when a first article is required (see 3.2 and 6.4). This inspection shall include the examination of 4.3 and the tests of 4.4. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the tests of 4.5, and the packaging inspection of 4.6.

4.3 Examination. Each sweeper shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 First article tests. When first article is required, the specified style and size (see 6.2), shall be tested as specified in 4.4.1 through 4.4.11, and when required 4.4.12. Failure to pass any phase of the required tests shall be cause for the Government to refuse acceptance of all sweepers until corrective action has been taken.

4.4.1 Operational tests. Operational tests for the sweeper shall cover all controls, electrical system, brushes, hopper dumping mechanism, vacuum system, brakes, and hydraulic system, when furnished. Sweeper shall be driven at all speeds, turns made at minimum radii, brakes tested for dependability, all controls, mechanisms, and movements tested to determine conformance to applicable requirements of this specification.

4.4.2 Sweeping test. The paved test area shall be thoroughly swept before test. A rectangular area, 50-feet long with a width 6 inches less than the maximum sweeping swath, shall be evenly covered with 1/2 lb of fine sand per square foot of area. Sweeper, with brush to ground contact set as recommended by the contractor, shall be operated in a one-way pass over the area. The sand remaining on the ground after the test shall be swept up, weighted, and compared with the total weight of sand used in the test to determine compliance with sweeping requirements of 3.8.

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4.4.3 Miscellaneous pickup test. Ten each of the following items shall be distributed over the area described in 4.4.2:

a. Wood, 1 by 1 by 6 inches.
b. Beverage cans, empty, 6 ounce (minimum). The size 28 sweeper shall be tested using paper drinking cups instead of beverage cans.
c. Nails, 2-1/2 inches long, common.
d. Wood chips, 1/4 by 1/4 by 1/2 inch.
e. Machine screws, 1/4-inch diameter by 2 inches long.
f. Hexagon nuts, 1/2-inch size.
g. Steel washers, 1-1/2 inch diameter by 1/8-inch thick.
h. Crumpled paper sheets (8-1/2 by 11 inches flat).
i. Aluminum sheet, 1/16 by 2 by 2 inches.
j. Twine (not less than 2-feet long).

Sweeper, with brush to ground contact set as recommended by the contractor, shall be operated over the area. The quantity of the items collected in the hopper shall be compared with the quantity distributed over the test area to determine compliance with 3.8.

4.4.4 Grade sweeping. Sweeper shall be operated in all directions on an included surface, covered under normally accumulated dirt and debris, to determine conformance to 3.8.

4.4.5 Dust control. A clean paved, rectangular area 50-feet long, with a width approximately equal to the swath of the main brush, shall be evenly covered with 1/2 bag (45 lb) of cement for sizes 28 and 36 sweepers, and one full bag (90 lb) of cement for sizes 42 and 48 sweepers. The hopper shall be empty, filters clean, main brush compartment skirt properly set to the floor, side brush raised, and main brush to ground contact set as recommended by contractor. The sweeper shall be driven over the area at the manufacturer's recommended speed and shall pick up cement in two passes without visible dust. No cleaning of the filter element shall be permitted during these tests. After picking up the cement, the sweeper shall be held stationary with the main brush and vacuum fan operating at th manufacturer's recommended speed. A smoke source, such as titanium tetrachloride on a cotton swab, shall be held near the bottom of the main brush housing at the rear, side, and front, approximately 3 inches from the housing. The vacuum system shall draw the smoke into the main brush compartment. The complete test specified herein shall be performed once for size 28 and 36 sweepers, and twice for size 42 and 48 sweepers without emptying the machine.

4.4.6 Hopper loading. Materials, consisting of normal dirt and debris as specified in 4.4.2, 4.4.3, and 4.4.5 amounting to 80 percent of the rated hopper capacity of the specified size sweeper (see 3.7), shall be evenly distributed over an area to be swept. All of the material shall be picked up and the sweeper driven not less than 100 feet to determine compliance with 3.13.11.3.

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4.4.7 Brake test. The fully loaded sweeper shall be driven down a 20 percent grade, dry, paved ramp at a travel speed of 5 mph. The service brakes shall be actuated and the distance from the point of application to stop point shall be measured. Sweeper shall again be driven down the ramp and stopped in the middle of the ramp. The parking brake shall be set and locked and the sweeper left parked for 30 minutes. This test shall be repeated with the sweeper facing the opposite direction. Test results shall comply with 3.13.7.

4.4.8 Obstruction test. Sweeper shall be operated over a fixed obstruction not less than 2-inches high, at least three times, at maximum sweeping speed specified in table I, to verify conformance to 3.8.

4.4.9 Noise level measurement. Sweeper shall be driven over a hard rough surface or a littered surface that imposes severe vibrations on the sweeper. Speeds shall be varied throughout the entire range. Sound level measurement at the operator's position shall be taken in accordance with SAE J919 to determine conformance to 3.11.

4.4.10 Reliability test. The unit shall be operated as a sweeper with brushes and vacuum dust control functioning at speeds from 0 to 5 mph for sizes 28, 36, and 42; and to 9 mph for size 48, for not less than 40 hours. At least 50 percent of the total time shall be for periods of not less than 3 hours continuous operation. At the conclusion of the test, the sweeper shall be subjected to and pass the sweeping tests specified in 4.4.2, 4.4.3, and 4.4.5. Failure of major components, excessive vibration and wear, or erratic operation shall be cause for rejection.

4.4.11 Lifting and tying down attachments. When furnished, lifting and tying down attachements shall be tested to determine conformance to 3.16. The manufacturer may furnish certified engineering calculations in lieu of a test to prove that the lifting and tying down attachments offered meets the requirements set forth in MIL-STD-209.

4.4.12 Test for electromagnetic interference characteristics. When electromagnetic interference characteristics are required, the sweeper shall be tested to determine conformance to 3.6.5. In lieu of tests to determine conformance to SAE J552, the manufacturer may prepare a certification that the sweeper meets the requirements, together with a list of the suppression devices installed. The list shall be sufficiently detailed to allow visual determination that the devices are installed.

4.5 Production unit tests. Each sweeper, after servicing and adjusting, shall be subjected to the operational tests specified in 4.4.1. Any evidence of nonconformance to the specified requirements shall be cause for rejection.

4.6 Preparation for delivery inspection. The preservation, packaging, packing, and marking of the item shall be inspected to verify conformance to the requirements of section 5.

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5. PREPARATION FOR DELIVERY

5.1 Vehicle processing. The equipment shall be preserved in accordance with the contractor's standard practice in a manner to prevent deterioration and damage. The equipment shall be prepared for shipment in a manner which will insure arrival at destination in a satisfactory condition. Preparation for delivery shall comply with applicable carrier rules and regulations.

6. NOTES

6.1 Intended use. Sweepers covered by this specification are intended for sweeping and cleaning warehouse aisles, shop and garage floors, and similar surface areas where fine cleaning and dust control is desired.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number, and date of this specification.
b. Type and size sweeper required (see 1.2).
c. When first article is required for inspection and approval (see 3.2, 4.4, and 6.3).
d. When cleaning, treatment, and painting shall conform to MIL-T-704, type A (see 3.6.1).
e. Color of finish coat required (see 3.6.1).
f. When treatment for fungus resistance is required (see 3.6.4).
g. When conformance to electromagnetic radiation suppression characteristics of SAE J552, or electromagnetic interference control requirements of MIL-STD-461 is required (see 3.6.5).
h. When lifting and tying down attachments are required (see 3.6.6).
i. Gasoline or diesel engine required (see 3.13.3.1).
j. When gasoline shall be other than that specified (see 3.13.3.2.1).
k. When battery is not required (see 3.13.3.2.3).
l. When special type of battery charging receptacle is required (see 3.13.3.3.4 and 6.4).
m. When tires shall be other than manufacturer's standard (see 3.13.6.2).
n. When the main brush shall be driven at constant speed (see 3.13.11.1).
o. Bristle material if other than specified (see 3.13.11.1 and 3.13.11.2).
p. Side brush requirement if other than as specified (see 3.13.11.2).
q. When the sweeper shall be equipped with both right and left side brushes (see 3.13.11.2).
r. When the sidebrushes shall be hydraulically driven (see 3.13.11.2).
s. When fire extinguisher is to be furnished and the type required (see 3.15).

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6.2.1 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

6.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Battery charging receptacle. When the receptacle or connector on the sweeper is to be compatible with an activities existing battery charging system, the specific type should be specified. Procurement of sweeper for the Post Office Department should specify receptacles and connectors manufactured by Anderson Power Products, 303 A Street, Boston, MA, as follows: for 36-V operation, a type N connector, catalog No. 4511A; for 24-V operation, a type SB receptacle, catalog No. 6313G1.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

MILITARY INTERESTS:

Custodians

Army - ME
Navy - YD
Air Force - 99

Review Activity

Air Force - 84

User Activities

Army - CE
DSA - CS

CIVIL AGENCY COORDINATING ACTIVITIES:

DOT - NHT
GSA - FSS
JUS - FPI
POSTAL - POS
VA - OSS

PREPARING ACTIVITY:

Navy - YD

Project No. 3825-0139

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein.

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